

The background features a collage of financial-related images: hands holding Euro banknotes, a wristwatch, and a document titled 'Borrowing Agreement'. These are overlaid with a black background containing a grid of white dots in the top right, a teal circle in the top left, and a green diagonal stripe. Four small green geometric shapes (circle, diamond, square, star) are in the top right corner.

BANK LOAN GRANTING CONSULTING

Presented By: Mohamed Amine KHALFALLAH – Frederic LU– Mohamed Sinofer VALIYA PEEDIYEKKAL

Introducing the Dataset

Number of Variables: 27



Numerical Variables:

- loan_amnt
- int_rate
- installment
- annual_inc
- dti
- open_acc
- pub_rec
- revol_bal
- revol_util
- total_acc
- mort_acc
- pub_rec_bankruptcies

Number of Rows: 396030



Categorical Variables:

- term
- grade
- sub_grade
- emp_title
- emp_length
- home_ownership
- verification_status
- loan_status
- purpose
- title
- initial_list_status
- application_type
- address



Date Variables:

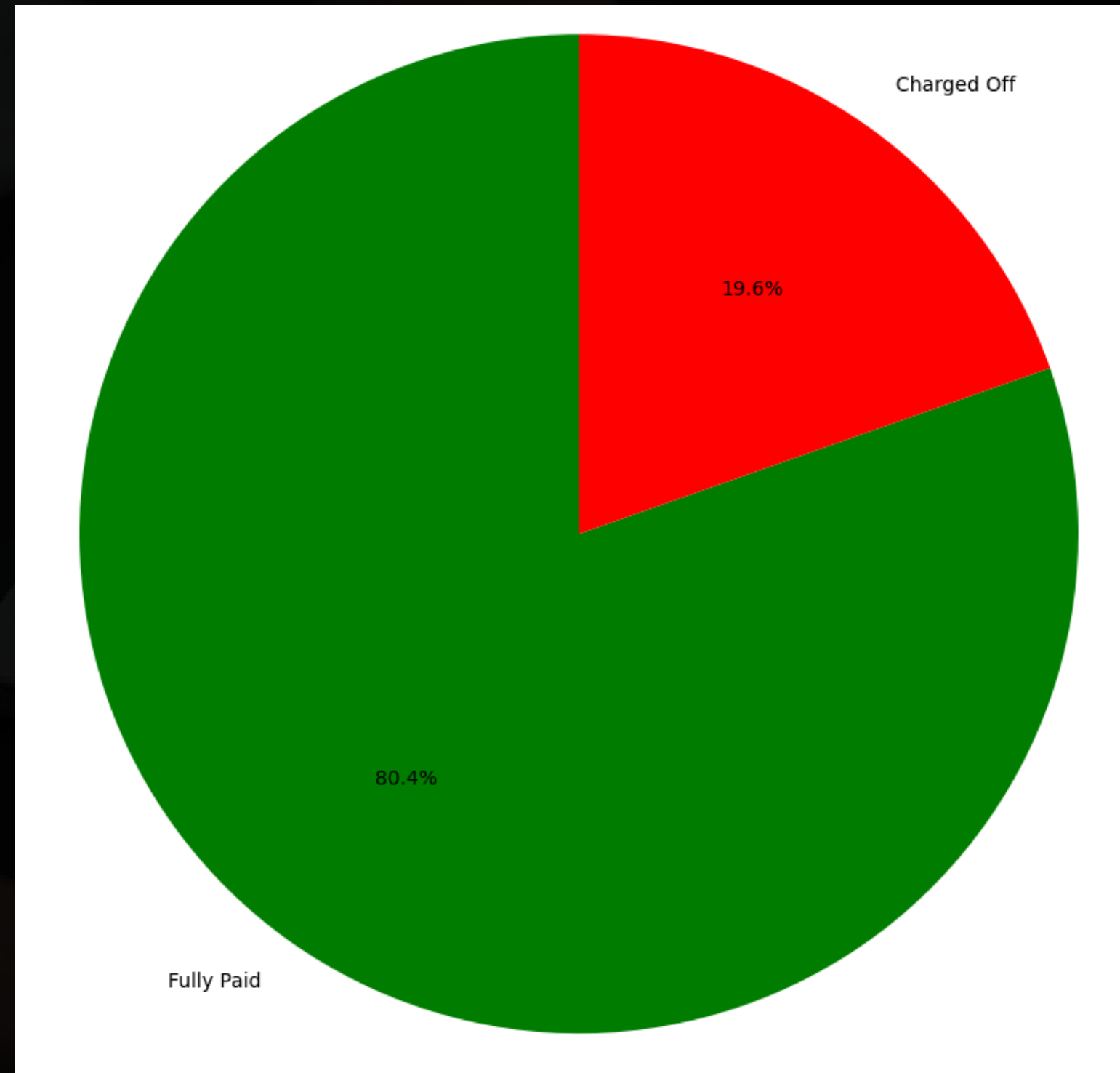
- issue_d
- Earliest_cr_line

Descriptive Analysis

	loan_amnt	int_rate	installment	annual_inc	dti	open_acc	pub_rec	revol_bal	revol_util
count	396030.000000	396030.000000	396030.000000	3.960300e+05	396030.000000	396030.000000	396030.000000	3.960300e+05	395754.000000
mean	14113.888089	13.639400	431.849698	7.420318e+04	17.379514	11.311153	0.178191	1.584454e+04	53.791749
std	8357.441341	4.472157	250.727790	6.163762e+04	18.019092	5.137649	0.530671	2.059184e+04	24.452193
min	500.000000	5.320000	16.080000	0.000000e+00	0.000000	0.000000	0.000000	0.000000e+00	0.000000
25%	8000.000000	10.490000	250.330000	4.500000e+04	11.280000	8.000000	0.000000	6.025000e+03	35.800000
50%	12000.000000	13.330000	375.430000	6.400000e+04	16.910000	10.000000	0.000000	1.118100e+04	54.800000
75%	20000.000000	16.490000	567.300000	9.000000e+04	22.980000	14.000000	0.000000	1.962000e+04	72.900000
max	40000.000000	30.990000	1533.810000	8.706582e+06	9999.000000	90.000000	86.000000	1.743266e+06	892.300000

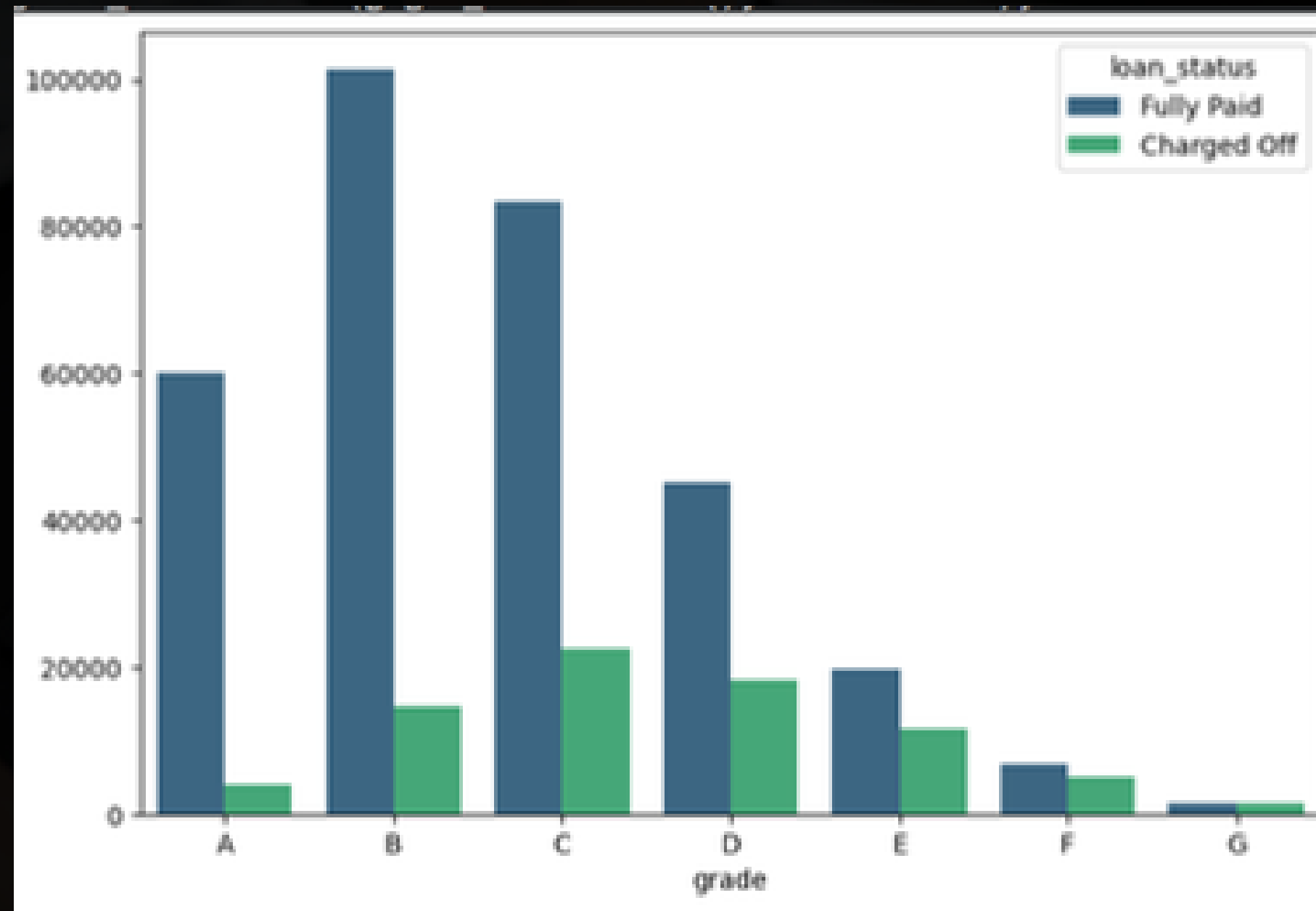
Descriptive Analysis

loan status Distribution



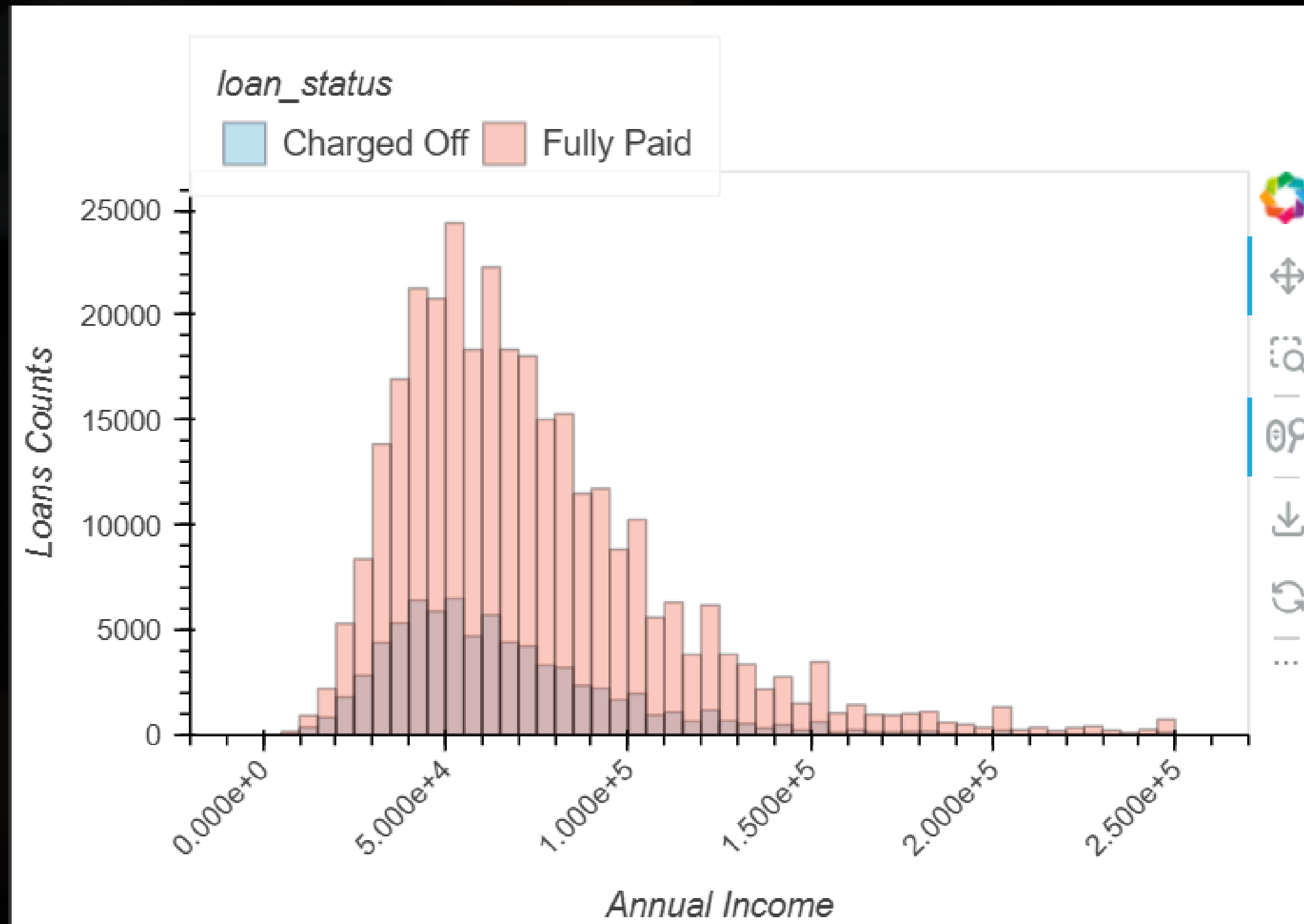
Descriptive Analysis

Loan status Distribution by grade



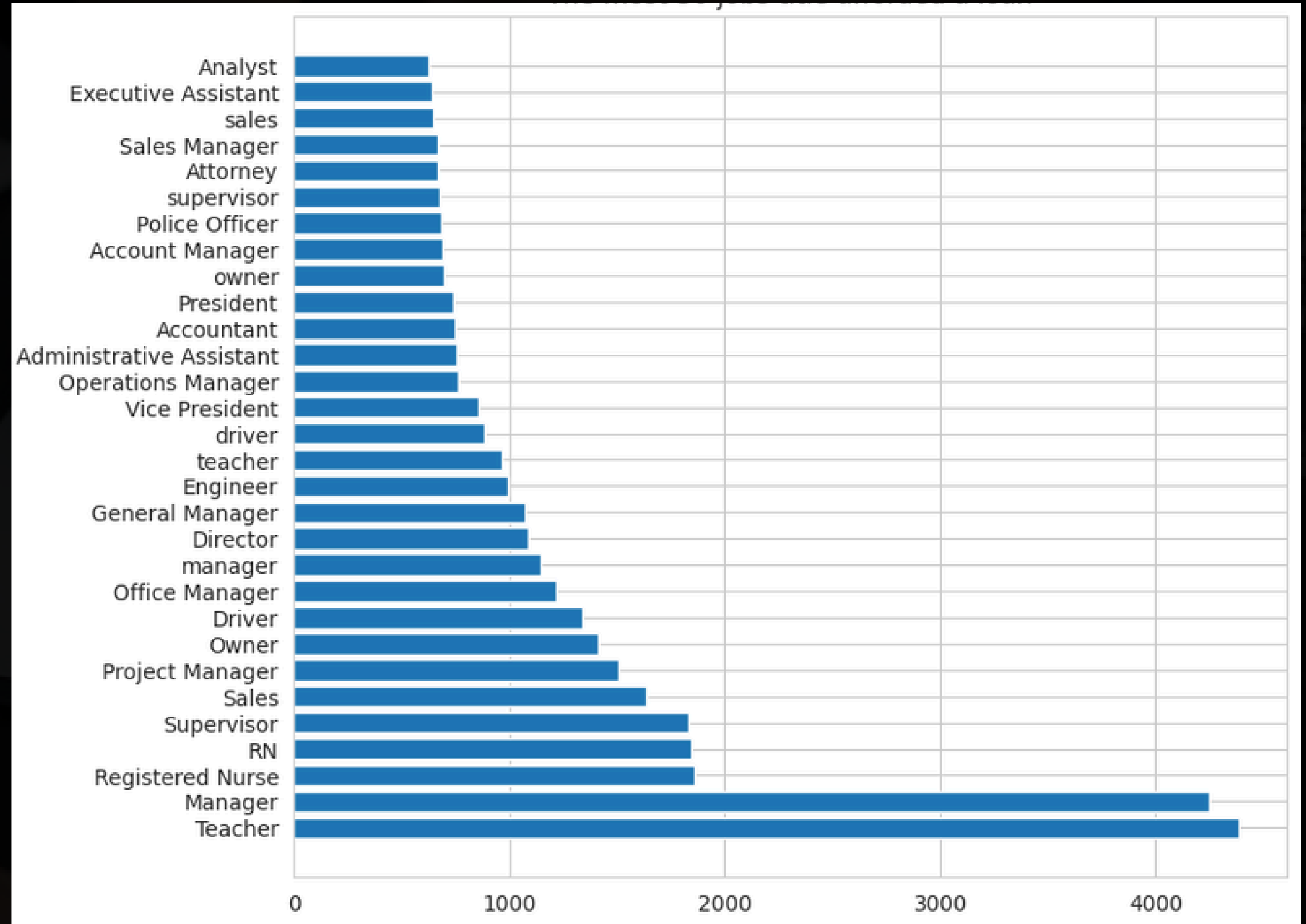
Descriptive Analysis

Loan status by Annual Income



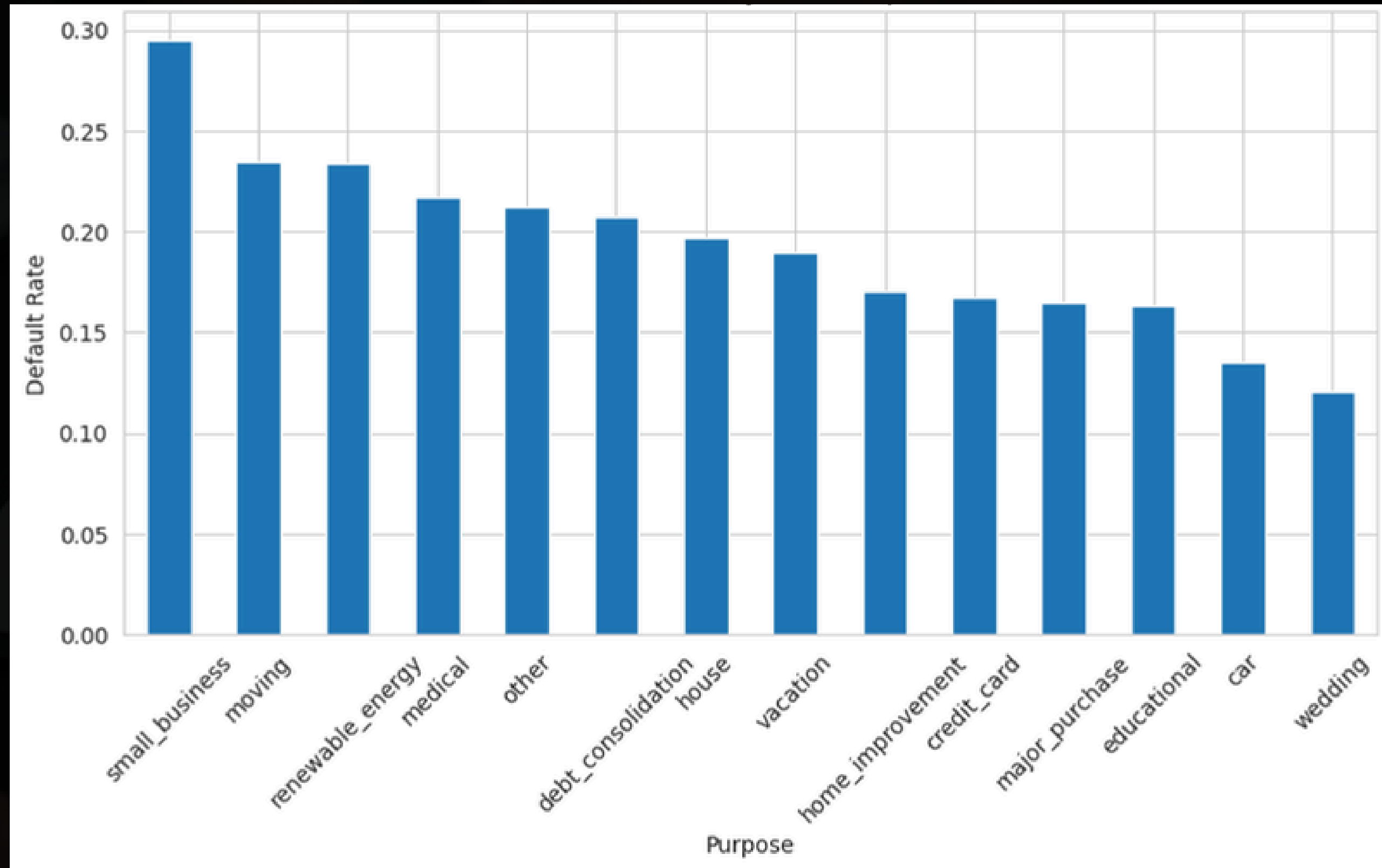
Descriptive Analysis

Number of Loans by Job Title

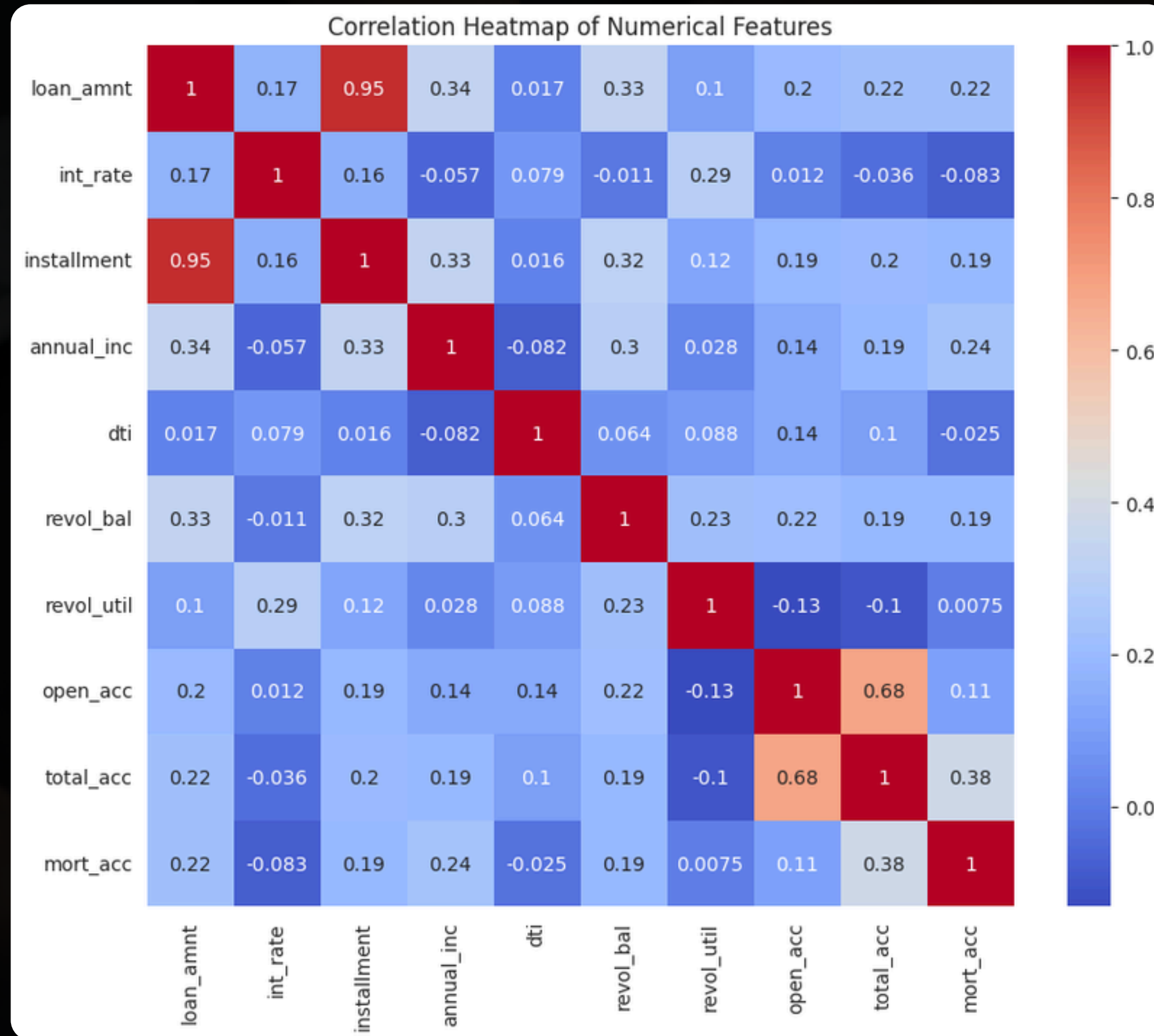


Descriptive Analysis

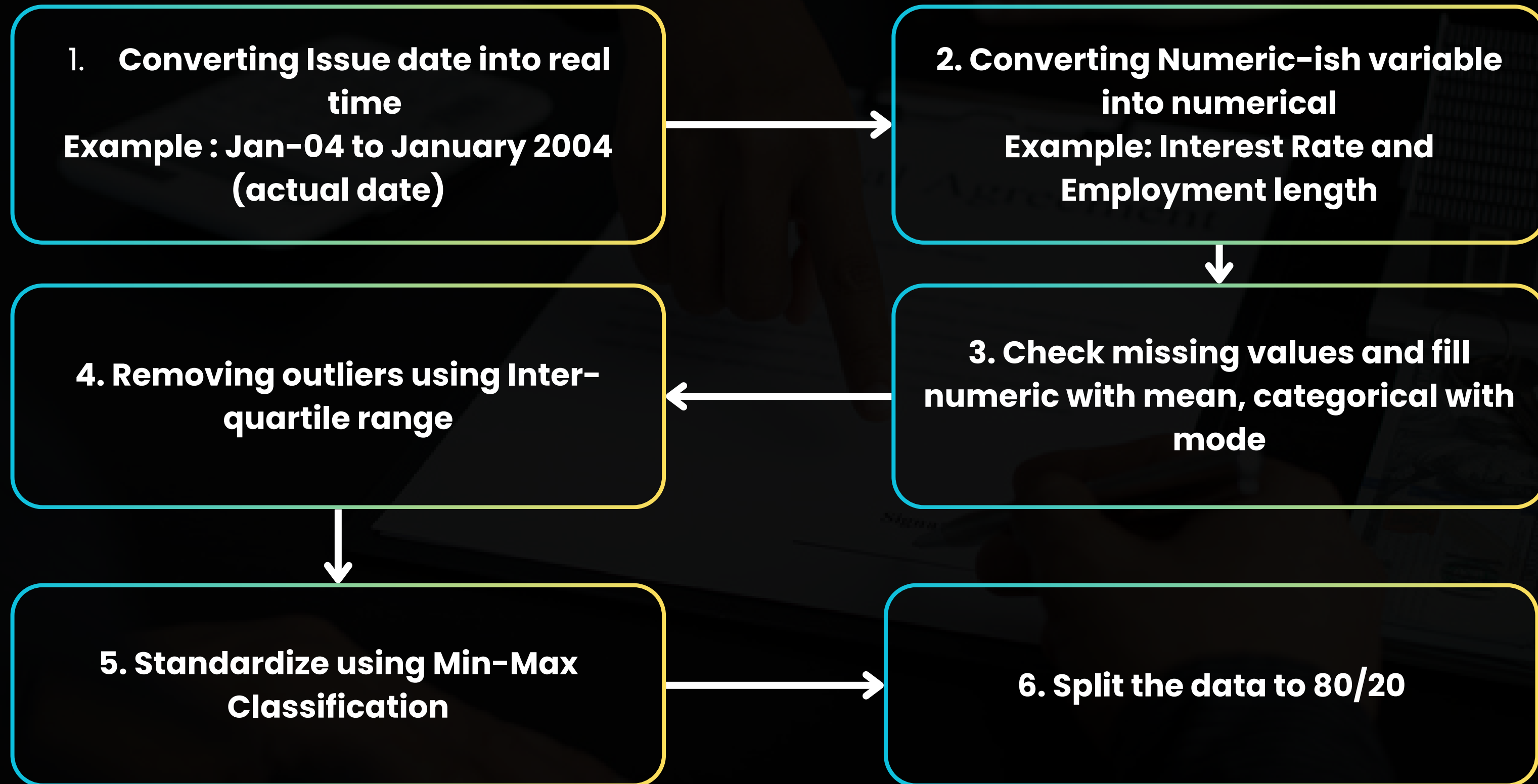
Default rate by loan purpose



Descriptive Analysis



DATA PREPROCESSING



Linear Regression Analysis

Model Methodology: Feature Engineering Strategy

Target variable: *Interest Rate*



1. Numeric features

We explicitly process four key continuous variables:

- **Loan Amount** (loan_amnt)
- **Employment Length** (emp_length_num)
- **Debt-to-Income** (dti)
- **Revolving Utilization** (revol_util)



2. Categorical features

We target two qualitative variables:

- **Purpose**: Reason for the loan.
- **Grade**: Credit quality rating.



3. Transformation

pd.get_dummies()
converts these categories into binary (0/1) columns.

drop_first=True
This removes the first category (Reference) to prevent multicollinearity

R SQUARED

0.907

RMSE

0.0650

INTERCEPT

0.0302

**Reference
variable**

- grade_A (Credit Score)
- purpose_car (Purpose)

VARIABLE	COEFFICIENT
const	0.0302
loan_amnt	0.0084
emp_length_num	0.0033
dti	0.0012
revol_util	0.3764
purpose_credit_card	0.0024
purpose_debt_consolidation	0.0062
purpose_educational	-0.0224
purpose_home_improvement	0.0050
purpose_house	0.0170
purpose_major_purchase	0.0026

purpose_medical	0.0100
purpose_moving	0.0124
purpose_other	0.0128
purpose_renewable_energy	0.0082
purpose_small_business	0.0110
purpose_vacation	0.0101
purpose_wedding	0.0175
grade_B	0.1835
grade_C	0.3338
grade_D	0.4883
grade_E	0.6256
grade_F	0.7804
grade_G	0.7650

Random Forest Result

ACCURACY

80.4%

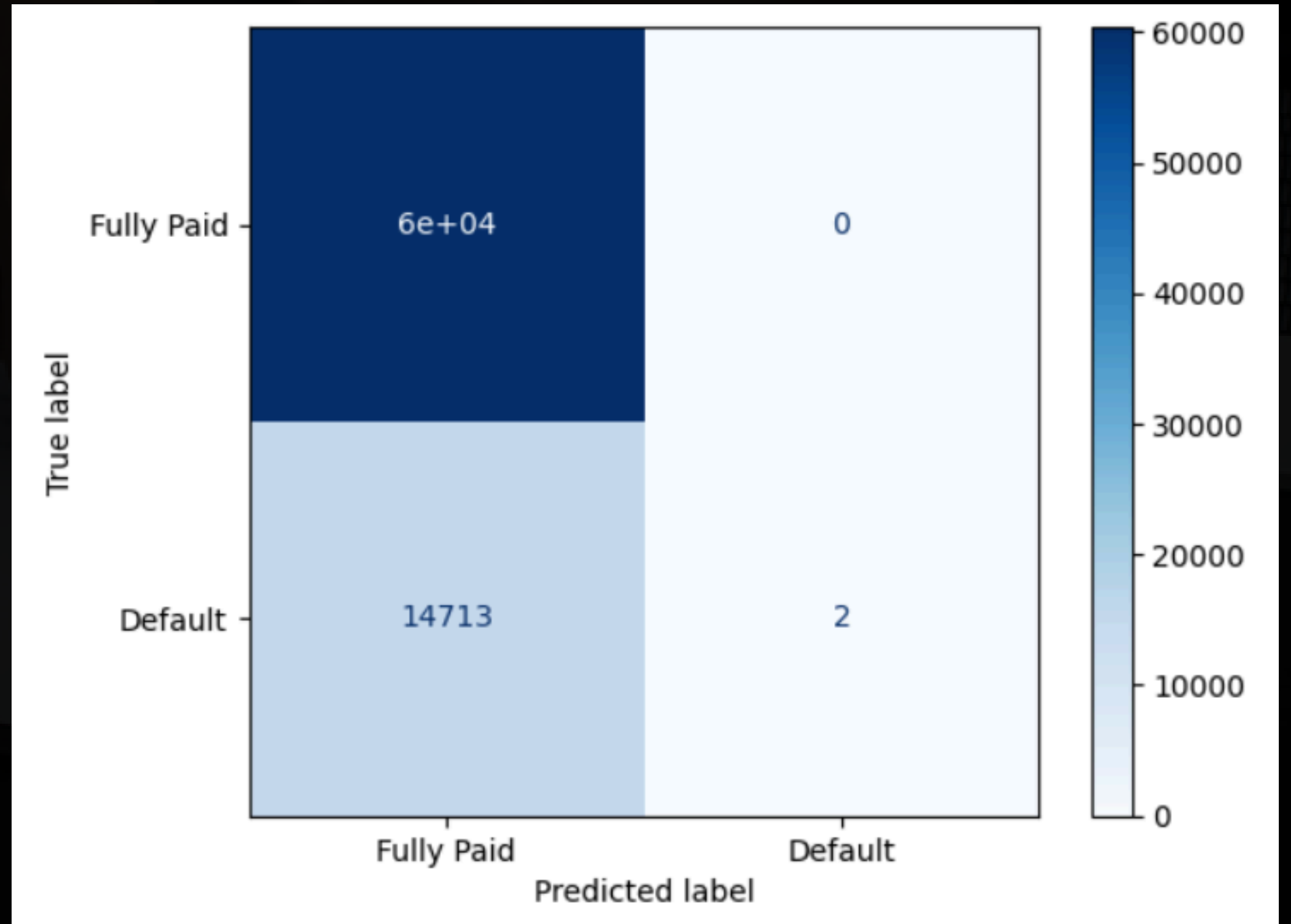
A model can achieve 80% accuracy simply by predicting "Fully Paid" for everything.

Fully Paid

- Precision: 0.80
- Recall: 1.00

Default

- Precision: 1.00
- Recall: 0.00



Random Forest Result

===== TOP 15 MOST IMPORTANT FEATURES =====

	Feature	Importance
29	grade_E	0.210452
3	dti	0.137084
28	grade_D	0.127311
30	grade_F	0.103578
26	grade_B	0.095347
2	annual_inc	0.067160
5	revol_util	0.054571
0	loan_amnt	0.052098
27	grade_C	0.036395
19	home_ownership_MORTGAGE	0.025183
23	home_ownership_RENT	0.020579
4	open_acc	0.014184
24	verification_status_Source Verified	0.011952
25	verification_status_Verified	0.011252
1	emp_length_num	0.008058

Ethics

Fairness and Bias

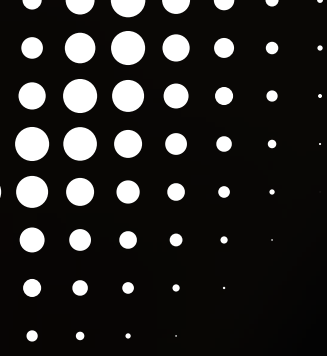
- **Problem:** Algorithmic Bias
- **Solution:** Treat every person exactly the same way when deciding on a loan.

Transparency and Explainability

- **Problem:** The "Black Box"
- **Solution:** Banks must use Explainable AI (XAI) techniques to pull apart the model's decision and provide a clear, easy-to-understand reason for an approval or denial.

Data Privacy and Security

- **Problem:** increasing the risk of privacy breaches
- **Solution:** Banks must follow strict privacy laws (like GDPR) to protect customer data.. This is done using techniques like federated learning or anonymization during AI training



How a similar company can use AI

Ethical & Explainable Lending (EEL)

- Ends the "Black Box"
- Self Checks for Unfairness
- Automated Audit Trail
- Approves the Right People



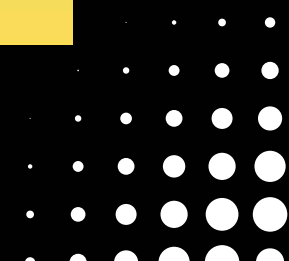
Dynamic Rate Optimization

- Maximize Profit per Loan
- Risk Assessment
- Win Ideal Customers
- Real-Time Adjustments



AI That Reads Loan Applications for Us

- Stop Guessing What a Loan is For
- Faster Approvals and Processing
- Cleaner Data for Better Decisions
- Catching Unusual Activity





THANK YOU
FOR YOUR
ATTENTION.

RMSE interpretation

Input Variables

TOTAL RATE RANGE

19.6273

(Max Rate - Min Rate)

NORMALIZED RMSE

0.0650

(Scaled Error 0-1)

1.28%

Real Average Prediction Error